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Nonuremic calciphylaxis in a patient with rheumatoid arthritis and osteoporosis treated with teriparatide

To the Editor: Calciphylaxis is a devastating condition in which metastatic calcification of the microvasculature results in thrombosis and ischemic necrosis of target tissues.¹⁻³ Calciphylaxis is well described in patients with chronic kidney disease and secondary hyperparathyroidism, however the literature describing it in the absence of these classic comorbidities is scant.

A 66-year-old woman presented with a 3-month history of painful lower extremity nodules. Her medical history included obesity, pulmonary emboli, osteoporosis, and a 20-year history of well-controlled rheumatoid arthritis. Medications included leflunomide, prednisone (2 mg daily), warfarin, and teriparatide [recombinant human

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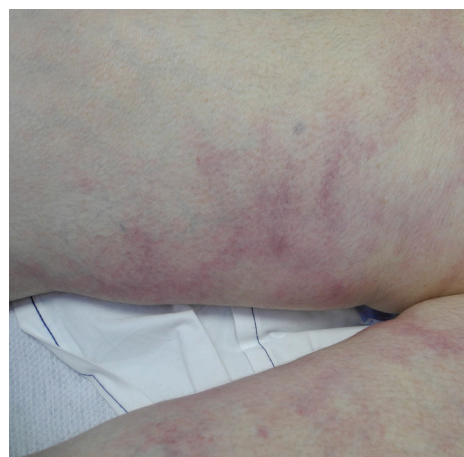


Fig 1. Nonuremic calciphylaxis in a patient treated with teriparatide: livedoid erythema and retiform purpura on the bilateral thighs.

parathyroid hormone (1-34)], initiated 5 months prior. Physical examination revealed a 10- × 3-cm indurated subcutaneous plaque with livedoid erythema and retiform purpura on the bilateral thighs and scattered indurated subcutaneous nodules on the bilateral lower extremities (Fig 1). A lateral thigh lesion was biopsied at an outside facility 2 months prior, leaving a nonhealing ulcer despite appropriate wound care. The biopsy specimen was consistent with thrombotic vasculopathy; however, no additional workup was performed. Von Kossa stains were ordered and read by our pathologist, but no changes of early calciphylaxis were seen. Laboratory studies demonstrated normal renal function, parathyroid hormone levels, and calcium-phosphate product. Serum and urine protein electrophoresis, antineutrophil cytoplasmic antibodies, antinuclear antibodies, and antiphospholipid antibodies were all negative.

Because of the patient's multiple risk factors, calciphylaxis was suspected clinically. Warfarin and teriparatide were discontinued and the patient was placed on low-molecular-weight heparin. Given her risk for poor wound healing, repeated biopsy was deferred. Bone scintigraphy of the thighs showed no abnormal uptake. Because of continued clinical suspicion and further lesion progression, the patient consented to repeated biopsy. Histopathology revealed intramural calcium deposition in subcutaneous arterioles with intimal hyperplasia and ischemic changes of the surrounding tissue consistent with calciphylaxis (Fig 2). Treatment with intravenous sodium thiosulfate therapy was initiated after surgical consultants determined the patient was a poor candidate for debridement. Unfortunately, her lesions progressed over the next month causing

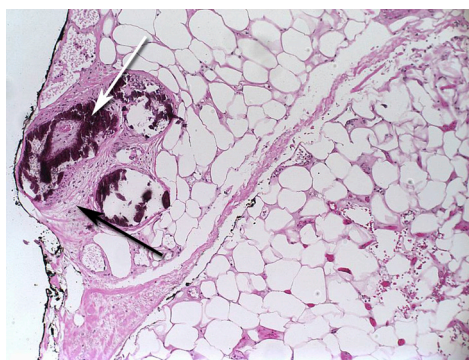


Fig 2. Nonuremic calciphylaxis in a patient treated with teriparatide: intramural calcium deposition (*white arrow*) in subcutaneous arterioles with intimal hyperplasia (*black arrow*) and ischemic changes of the surrounding tissue. (Hematoxylin-eosin stain; original magnification: $\times 100$.)

intractable pain despite management by pain specialists. Ultimately, she was transitioned to hospice care and died soon thereafter.

This case underscores the importance of high clinical suspicion and prompt diagnosis in patients presenting with livedoid erythema, painful indurated subcutaneous plaques, and ulcerations in adipose-rich areas who have risk factors for nonuremic calciphylaxis including female gender, obesity, chronic inflammation, liver disease, rapid weight loss, malignancy, and primary hyperparathyroidism.¹ Corticosteroids,¹ warfarin,¹ and teriparatide² have all been implicated as well. These factors are all thought to disrupt the balance between nuclear factor-kappa beta (NF κ B)-mediated activation and fetuin-A and matrix Gla-protein-mediated (MGP) inhibition of endothelial cell calcification.³ Because of the patient's multiple risk factors for calciphylaxis, the relationship to teriparatide cannot be stated with certainty, however previous cases exist.^{2,4}

Calciphylaxis may be diagnosed clinically in patients with typical skin findings and multiple risk factors. Because histopathology is often nondiagnostic in calciphylaxis, the benefits of a biopsy must be weighed against the risk of poor wound healing. If biopsy is performed, special stains may aid in diagnosis as histopathologic changes of early calciphylaxis are subtle.⁵ Clearer definition of diagnostic criteria for calciphylaxis is needed.

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Alcohol-induced vasculitis: Case report and commentary

To the Editor: A 26-year-old woman with no significant medical history on no medications or supplements was evaluated for a 2-year history of episodic tender, nonpruritic, purpuric macules and papules on the back of her feet and legs with associated edema. She reported that several hours after drinking beer, wine, or liquor, she develops purpuric lesions (*Fig 1*) that resolved within 2 to 7 days without sequelae. She had experienced at least 8 episodes, each directly linked to alcohol ingestion. She had no skin lesions in the 6 months before evaluation during which she abstained from drinking any alcohol.

A provocation test consisted of drinking 8 oz of alcohol. Within 6 hours she developed subtle pink macules and papules on her legs. Eight hours after ingestion, 1 of these lesions was biopsied and revealed a perivascular neutrophilic infiltrate and leukocytoclasia with focal vascular wall damage and fibrinoid changes (*Fig 2*). On direct immunofluorescence, 1 small superficial dermal vessel demonstrated granular perivascular staining with IgA (not shown).

Four cases of alcohol-associated purpura have been reported in the literature¹⁻⁴; none of these cases included biopsy evidence of vascular wall damage. Alibrandi et al¹ reported a case of a 60-year-old man with recurrent asymptomatic